7	1.0	−1.0	−1.0	−1.0	0.0	1.5	0.0	1.0	0.0	0.0
8	0.0	−1.5	−0.5	1.5	0.0	0.0	0.0	−1.0	0.5	−0.5
9	−2.0	−1.5	1.5	1.5	0.0	0.0	0.5	1.0	0.0	1.0
10	−0.5	3.5	0.0	−1.0	−1.5	−1.5	−1.0	−1.0	1.0	0.5
11	0.0	1.5	0.0	0.0	2.0	−1.5	0.5	−0.5	2.0	−1.0
12	0.0	−2.0	−0.5	0.0	−0.5	2.0	1.5	0.0	0.5	−1.0
13	−1.0	−0.5	−0.5	−1.0	0.0	0.5	0.5	−1.5	−1.0	−1.0
14	0.5	1.0	−1.0	−0.5	−2.0	−1.0	−1.5	0.0	1.5	1.5
15	1.0	0.0	1.5	1.5	1.0	−1.0	0.0	1.0	−2.0	−1.5

(c) Set up an s^2 chart and compare with the s chart in part (a).

6.8. Samples of $n = 6$ items each are taken from a process at regular intervals. A quality characteristic is measured, and $\bar{x}$ and R values are calculated for each sample. After 50 samples, we have

$$\sum_{i=1}^{50} \bar{x}_i = 2000 \quad \text{and} \quad \sum_{i=1}^{50} R_i = 200$$

Assume that the quality characteristic is normally distributed.

(a) Compute control limits for the $\bar{x}$ and R control charts.

(b) All points on both control charts fall between the control limits computed in part (a). What are the natural tolerance limits of the process?

(c) If the specification limits are 41 ± 5.0, what are your conclusions regarding the ability of the process to produce items within these specifications?

(d) Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process producing?

74.000 + 0.05 mm.

(a) Set up $\bar{x}$ and R co
the process in stat

(b) Note that the cor
(a) are identical
chart in Exam
based on s. Wi

(c) Estimate proc
process. Estin
produced that

6.11. Table 6E.6 show
ton ring process
control charts w
$\bar{x}$ and R chart
process in con

■ TABLE
Piston Ring D

Sample Number, i	
26	74.01
27	73.99
28	73.9
29	74.0
30	74.(
31	73.
32	74
33	74
34	7
35	7
36	
37	
38	
39	
40	

(e) Make suggestions as to how the process performance could be improved.

6.9. Control charts on $\bar{x}$ and s are to be maintained on the torque readings of a bearing used in a wingflap actuator assembly. Samples of size $n = 10$ are to be used, and we know from past experience that when the process is in control, bearing torque has a normal distribution with mean $\mu = 80$ inch-pounds and standard deviation $\sigma = 10$ inch-pounds. Find the center line and control limits for these control charts.

6.10. Consider the piston ring data shown in Table 6E.1. Assume that the specifications on this component are 74.000 + 0.05 mm.

(a) Set up $\bar{x}$ and R control charts on this process. Is the process in statistical control?

(b) Note that the control limits on the $\bar{x}$ chart in part (a) are identical to the control limits on the $\bar{x}$ ... Example 6.1 where the limits were

(a) Establish $\overline{x}$ and R control charts for compressive strength using these data. Is the process in statistical control?

(b) After establishing the control charts in part (a), 15 new subgroups were collected and the compressive strengths are shown in Table 6E.8. Plot the $\overline{x}$ and R values against the control units from part (a) and draw conclusions.

6.14. Reconsider the data presented in Exercise 6.13.

(a) Rework both parts (a) and (b) of Exercise 6.13 using the $\overline{x}$ and s charts.

(b) Does the s chart detect the shift in process variability more quickly than the R chart did originally in part (b) of Exercise 6.13?

S7 2018 Q.5

6.15. Samples of $n = 4$ items are taken from a process at regular intervals. A normally distributed quality characteristic is measured and $\overline{x}$ and s values are calculated for each sample. After 50 subgroups have been analyzed, we have

$$\sum_{i=1}^{50} \overline{x}_i = 1000 \quad \text{and} \quad \sum_{i=1}^{50} s_i = 72$$

(a) Compute the control limits for the $\bar{x}$ and s control charts.

(b) Assume that all points on both charts plot within the control limits. What are the natural tolerance limits of the process?

(c) If the specification limits are 19 ± 4.0, what are your conclusions regarding the ability of the process to produce items conforming to specifications?

(d) Assuming that if an item exceeds the upper specification limit it can be reworked, and if it is below the lower specification limit it must be scrapped, what percent scrap and rework is the process now producing?

(e) If the process were centered at $\mu = 19.0$, what would be the effect on percent scrap and rework?

6.16. Consider the $\bar{x}$ and R chart that you established in Exercise 6.10 for the piston ring process. Suppose that you want to continue control charting piston ring diameter using $n = 3$. What limits would be used on the $\bar{x}$ and R chart?

6.17. Samples of size $n = 5$ are collected from a process every half hour. After 50 samples have been collected, we calculate $\bar{\bar{x}} = 20.0$ and $\bar{s} = 1.5$. Assume that both charts exhibit control and that the quality characteristic is normally distributed.

(a) Estimate the process standard deviation.

(b) Find the control limits on the $\bar{x}$ and s charts.

(c) If the process mean shifts to 22, what is the probability of concluding that the process is still in control?

6.18. Samples of size $n = 5$ are taken from a manufacturing process every hour. A quality characteristic is measured, and $\bar{x}$ and R are computed for each sample. After 25 samples have been analyzed, we have

$$\sum_{i=1}^{25} \bar{x}_i = 662.50 \quad \text{and} \quad \sum_{i=1}^{25} R_i = 9.00$$

The quality characteristic is normally distributed.

(a) Find the control limits for the $\bar{x}$ and R charts.

(b) Assume that both charts exhibit control. If the specifications are 26.40 ± 0.50, estimate the fraction nonconforming.

(c) If the mean of the process were 26.40, what fraction nonconforming would result?

6.19. Control charts for $\bar{x}$ and R are maintained for an important quality characteristic. The sample size is $n = 7$; $\bar{x}$ and R are computed for each sample. After 35 samples, we have found that

$$\sum_{i=1}^{35} \bar{x}_i = 7805 \quad \text{and} \quad \sum_{i=1}^{35} R_i = 1200$$

(a) Set up $\bar{x}$ and R charts using these data.

(b) Assuming that both charts exhibit control, estimate the process mean and standard deviation.

(c) If the quality characteristic is normally distributed and if the specifications are 220 ± 35, can the process meet the specifications? Estimate the fraction nonconforming.

(d) Assuming the variance to remain constant, state where the process mean should be located to minimize the fraction nonconforming. What would be the value of the fraction nonconforming under these conditions?

6.20. A process is to be monitored with standard values $\mu = 10$ and $\sigma = 2.5$. The sample size is $n = 2$.

(a) Find the center line and control limits for the $\bar{x}$ chart.

(b) Find the center line and control limits for the R chart.

(c) Find the center line and control limits for the s chart.

6.21. A critical dimension of a machined part has specifications 100 ± 10. Control chart analysis indicates that the process is in control with $\bar{\bar{x}} = 104$ and $\bar{R} = 9.30$. The control charts use samples of size $n = 5$. If we assume that the characteristic is normally distributed, can the mean be located (by adjusting the tool position) so that all output meets specifications? What is the present capability of the process?

6.22. An $\bar{x}$ chart is used to control the mean of a normally distributed quality characteristic. It is known that $\sigma = 6.0$ and $n = 4$. The center line = 200, UCL = 209, and LCL = 191. If the process mean shifts to 188, find the probability that this shift is detected on the first subsequent sample.

6.23. A TiW layer is deposited on a substrate using a sputtering tool. Table 6E.9 contains layer thickness measurements (in Angstroms) on 20 subgroups of four substrates.